COVID-19 TIMESERIES FORECASTING

on vaccination data with deep learning

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4/19/21

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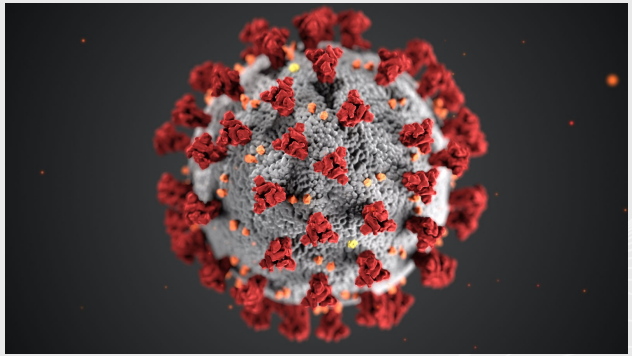


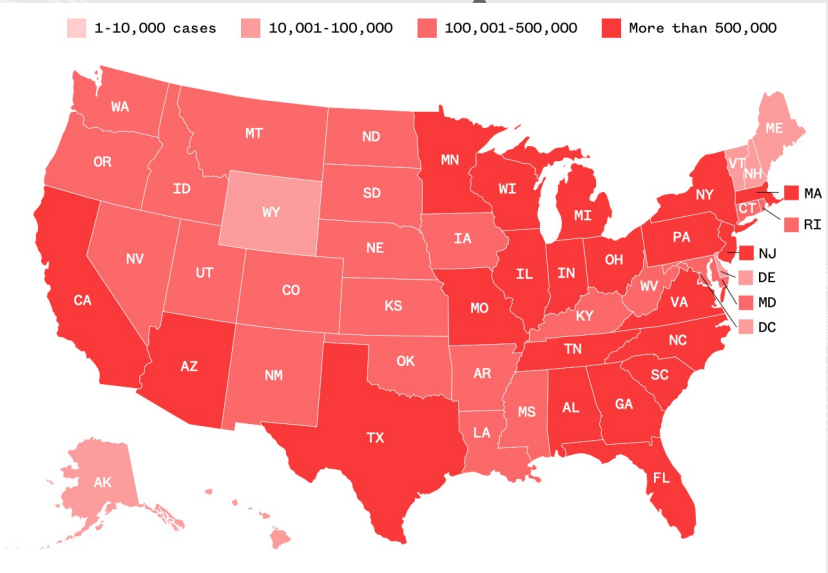
01.

INTRODUCTION

INTRODUCTION

The COVID-19 pandemic prevails as an ultimatum to the global economic growth and the wellbeing of society

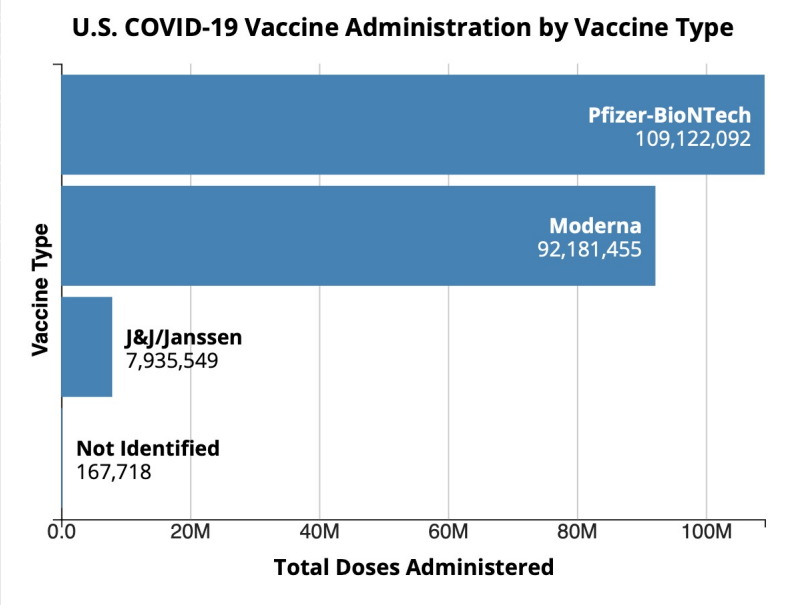
The global spread of COVID-19 is increasing day by day, creating a larger risk of disease or death as well as a strain on the economy[1]



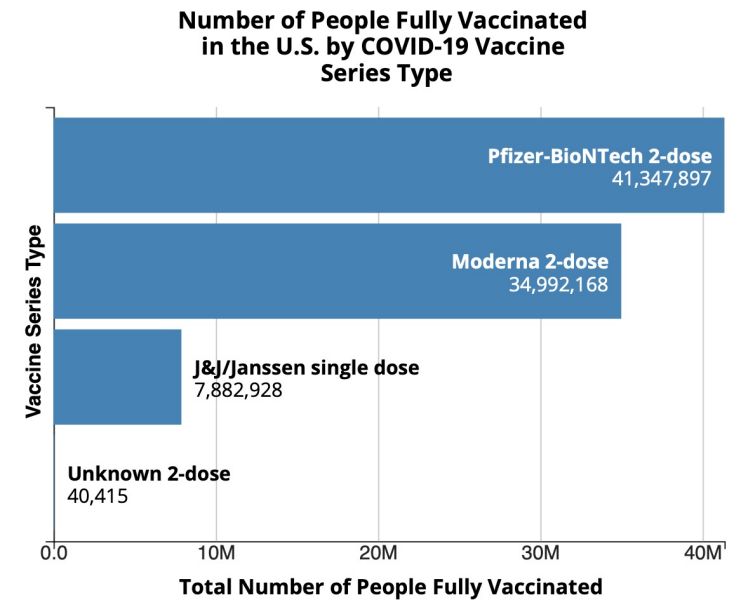
Vaccines!

There are currently three vaccines that have received emergency authorization from the FDA[2]

- Pfizer
- Moderna
- Johnson & Johnson*



Total doses administered as of 4/18/21[1]



Number of people fully vaccinated as of 4/18/21[1]

Massachusetts Reopening Plan

Current state: Stay at home	Phase 1: Start	Phase 2: Cautious	Phase 3: Vigilant	Phase 4: New normal
Essential business and services only	Limited industries resume operations with severe restrictions	Additional industries resume operations with restrictions and capacity limitations	Additional industries resume operations with guidance	Development of vaccines and / or treatments enable resumption of "new normal"

Can two weeks worth of daily covid vaccination data be predicted with <u>80%</u> accuracy?

02.

Methodology

Methodology

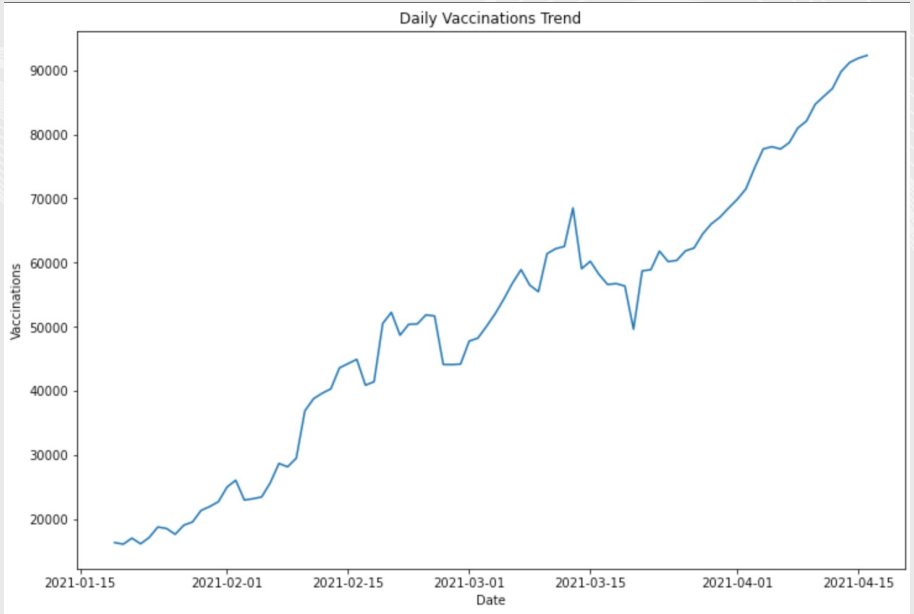
Data Import
Pulled from “Our World in Data” github repository[4]

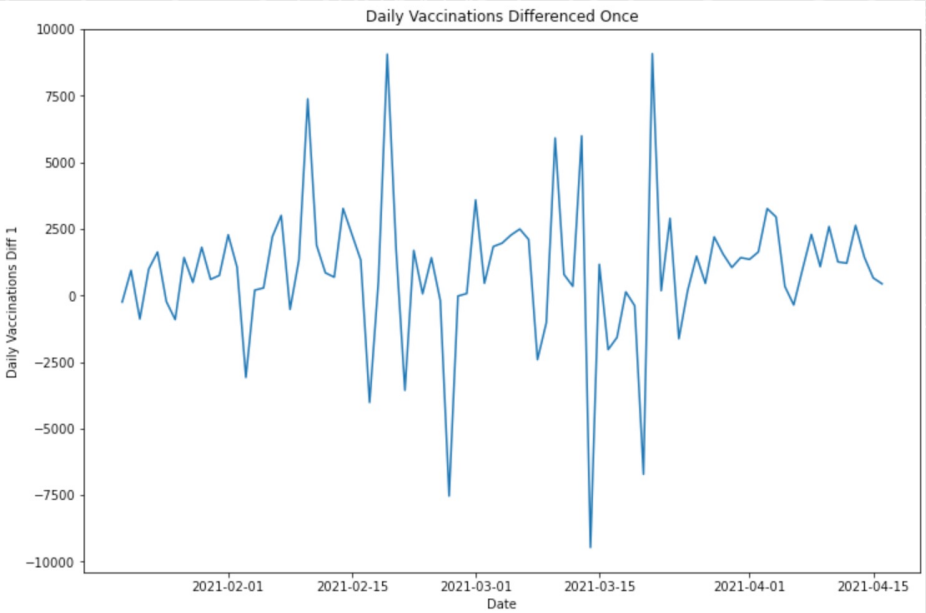
EDA
Data checked for missing values, autocorrelation, stationarity

Pre-processing
Data set up in three dimensions to be processed by RNN

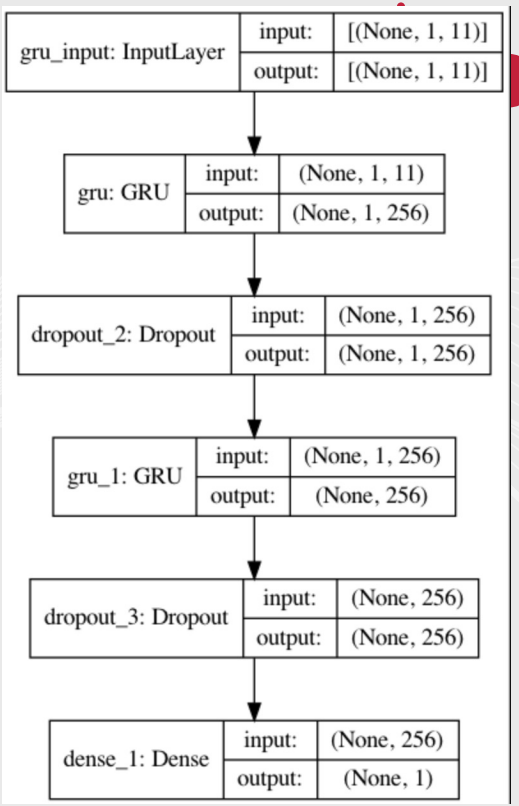
Models
SimpleRNN,
GRU,
LSTM

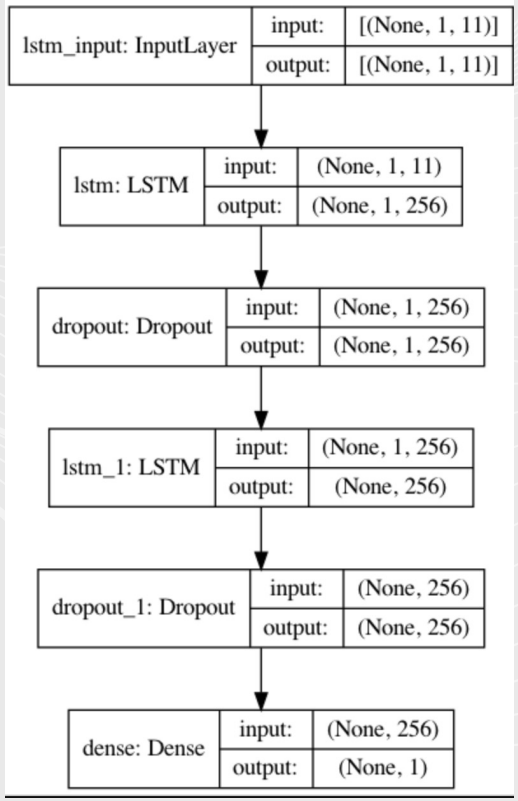
Daily Vaccination Stationarity

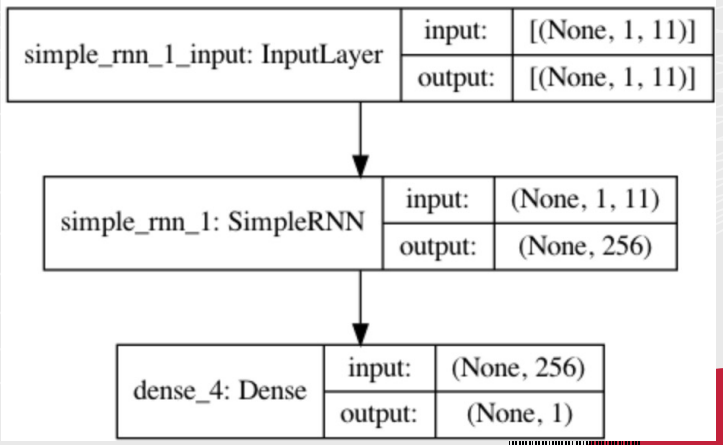




03. Results

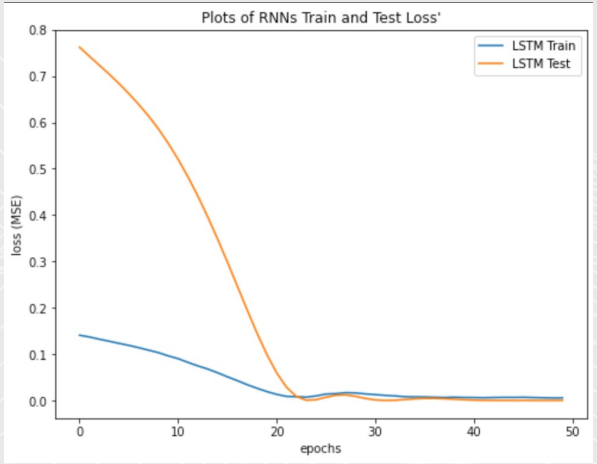

gru_input: InputLayer
input: [(None, 1, 11)]
output: [(None, 1, 11)]
gru: GRU
input: (None, 1, 11)
output: (None, 1, 256)
dropout_2: Dropout
input: (None, 1, 256)
output: (None, 1, 256)
gru_1: GRU
input: (None, 1, 256)
output: (None, 256)
dropout_3: Dropout
input: (None, 256)
output: (None, 256)
dense_1: Dense
input: (None, 256)
output: (None, 1)

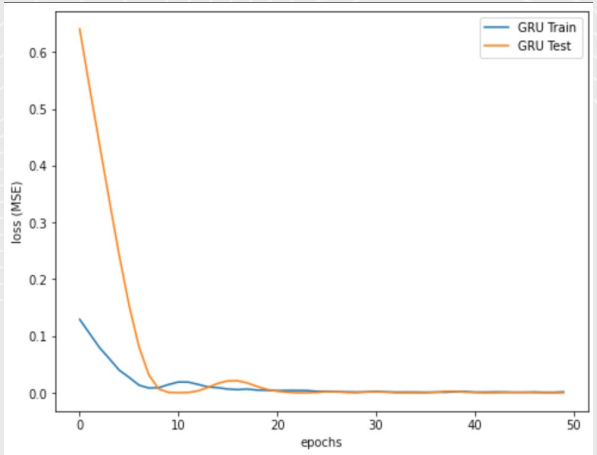

lstm_input: InputLayer
input: [(None, 1, 11)]
output: [(None, 1, 11)]
lstm: LSTM
input: (None, 1, 11)
output: (None, 1, 256)
dropout: Dropout
input: (None, 1, 256)
output: (None, 1, 256)
lstm_1: LSTM
input: (None, 1, 256)
output: (None, 256)
dropout_1: Dropout
input: (None, 256)
output: (None, 256)
dense: Dense
input: (None, 256)
output: (None, 1)


simple_rnn_1_input: InputLayer
input: [(None, 1, 11)]
output: [(None, 1, 11)]
simple_rnn_1: SimpleRNN
input: (None, 1, 11)
output: (None, 256)
dense_4: Dense
input: (None, 256)
output: (None, 1)

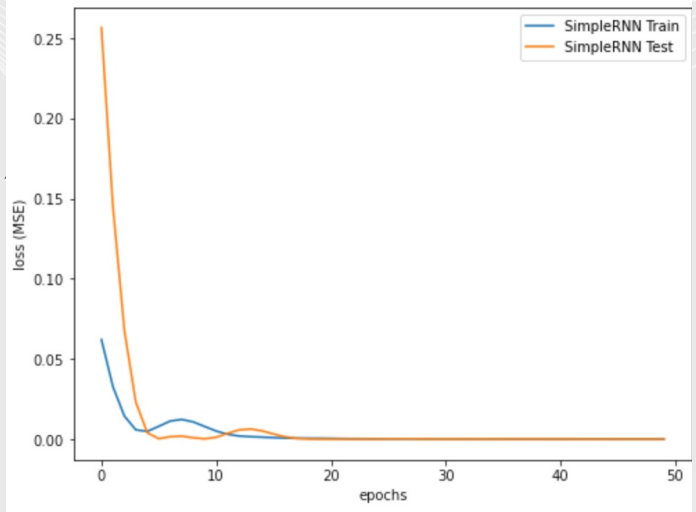
Mean Squared Error

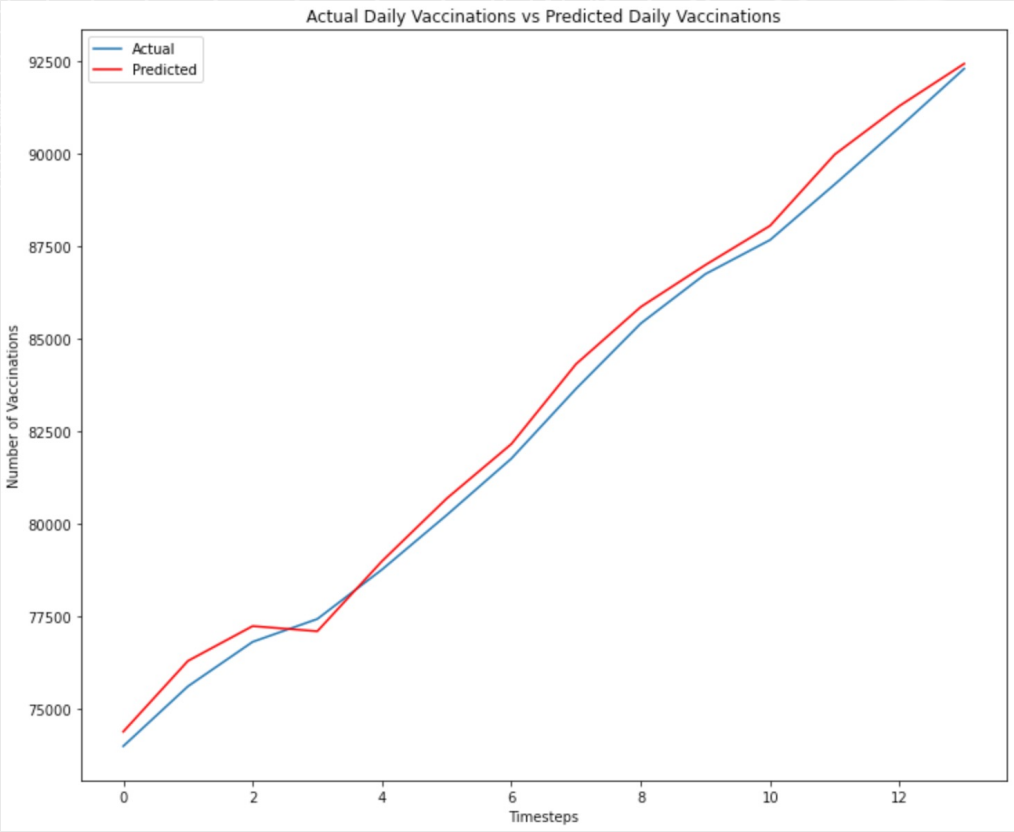
	Train MSE	Test MSE
LSTM	27,325,565	1,947,489
GRU	1,385,403	449,024
SimpleRNN	691,060	227,063



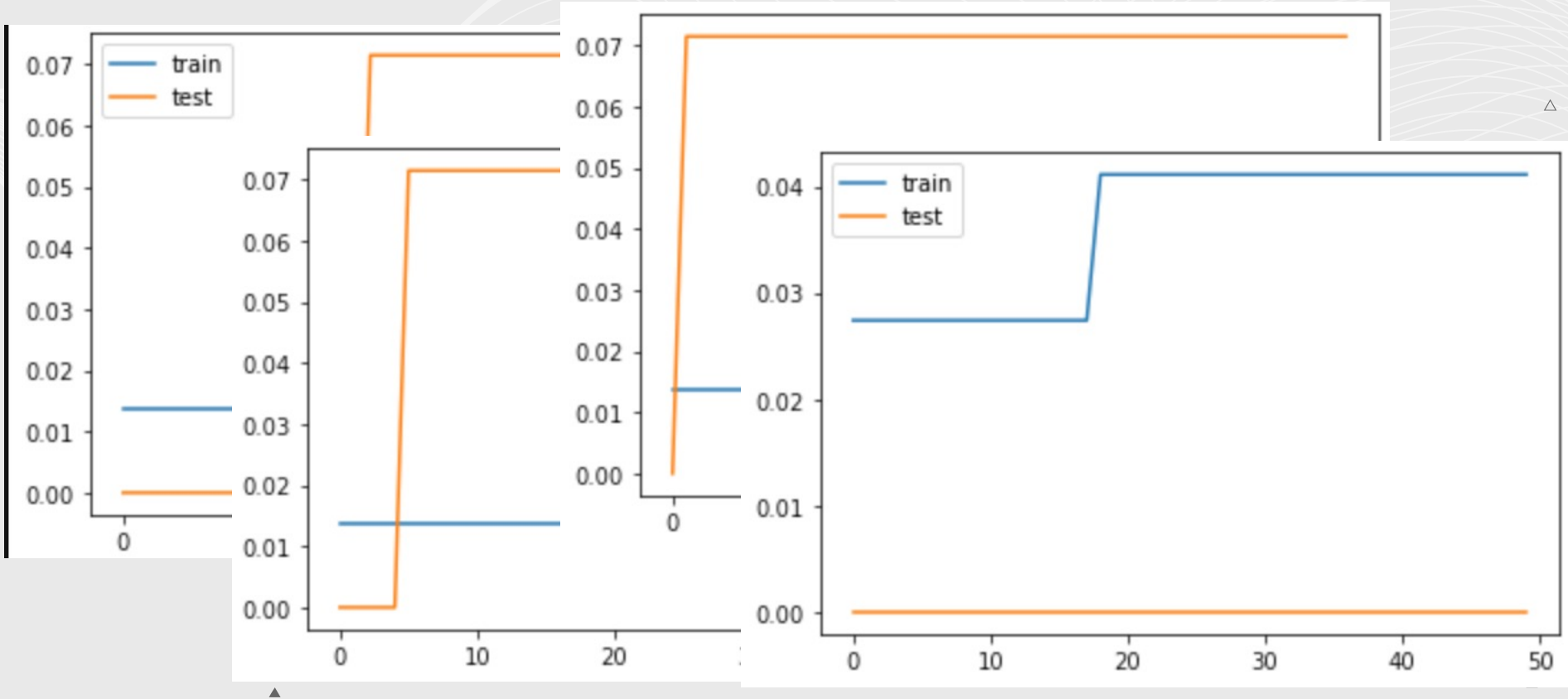


SimpleRNN 14-day Prediction





What about the accuracy?



04.

Conclusion

Conclusions

- RNNs can be used to perform timeseries forecasting on vaccination data
- SimpleRNN worked the best, most likely because of the size of the data set

Future Work

- Figure out accuracy!
- Feature selection
- Look at other parameters/optimizers
- Website that updates daily to visualize predictions at a county level

Citations

1. https://covid.cdc.gov/covid-data-tracker/#datatracker-home
2. https://www.fda.gov/vaccines-blood-biologics/vaccines/emergency-use-authorization-vaccines-explained
3. https://www.mass.gov/info-details/reopening-massachusetts
4. https://github.com/owid/covid-19-data

THANKS!

Do you have any questions?

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